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Pre-Lab: Undersatnading Google Collab and Python librariers for machine learning.

1. Numpy
2. Pandas
3. Matplotlib
4. Seaborn
5. Sckit-Learn

Double-click (or enter) to edit

Numpy

```
import numpy as np
#Create Array
a = np.array([1,2,3,4])
b = np.zeros((2,3))
c = np.ones((2,3))
d = np.random.rand(2,3)
print(a)
print(b)
print(c)
print(d)
```

```
[1 2 3 4]
[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]]
[[0.84706224 0.68156768 0.63311236]
 [0.5828139  0.48905036 0.74126882]]
```

```
#Array Operations
print(a.shape)
print(a.mean())
print(a.std())
print(np.sum(a))
print(np.max(a))
```

```
(4,)
2.5
1.118033988749895
10
4
```

```
#Matrix Operations
x = np.array([[1,2],[3,4]])
y = np.array([[5,6],[7,8]])
print(np.dot(x,y))
print(np.transpose(x))
```

```
[[19 22]
 [43 50]]
[[1 3]
 [2 4]]
```

Pandas

```
import pandas as pd
#Create Dataframe
data = {
    'Age': [25,30,35,40],
    'Salary' : [30000,50000,70000,90000],
    'Department' : ['IT','HR','IT','Finance']
}
df = pd.DataFrame(data)
print(df.head())
print(df.info())
print(df.describe())
print(df['Salary'])
print(df[['Age','Department']])
```

```
   Age  Salary Department
0   25   30000         IT
1   30   50000         HR
2   35   70000         IT
3   40   90000    Finance
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Age         4 non-null      int64
 1   Salary      4 non-null      int64
 2   Department  4 non-null      object
dtypes: int64(2), object(1)
memory usage: 228.0+ bytes
None
             Age        Salary
count   4.000000      4.000000
mean   32.500000  60000.000000
std     6.454972  25819.888975
min    25.000000  30000.000000
25%    28.750000  45000.000000
50%    32.500000  60000.000000
75%    36.250000  75000.000000
max    40.000000  90000.000000
0    30000
1    50000
2    70000
3    90000
Name: Salary, dtype: int64
   Age Department
0   25         IT
1   30         HR
2   35         IT
3   40    Finance
```

```
df['Salary'].fillna(df['Salary'].mean(),inplace=True)
print(df)
```

```
   Age  Salary Department
0   25   30000         IT
1   30   50000         HR
2   35   70000         IT
3   40   90000    Finance
```

```
df_encoded = pd.get_dummies(df,columns=['Department'])
print(df_encoded)
```

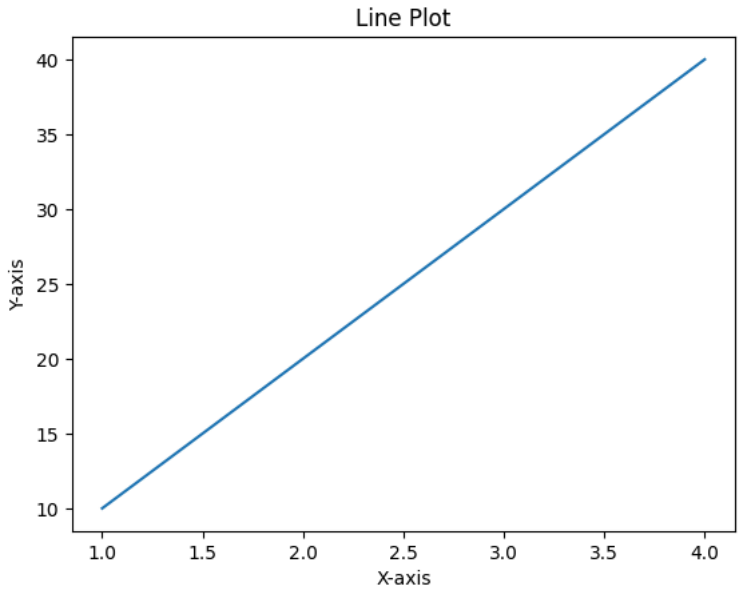
```
   Age  Salary  Department_Finance  Department_HR  Department_IT
0   25   30000               False          False           True
1   30   50000               False           True          False
2   35   70000               False          False           True
3   40   90000                True          False          False
```

Matplotlib

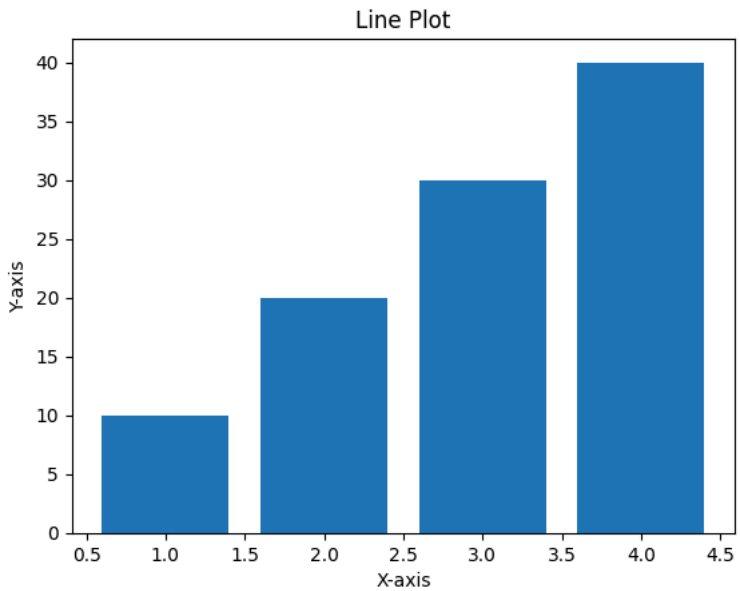
```
import matplotlib.pyplot as plt
```

```
x = [1,2,3,4]
y = [10,20,30,40]
```

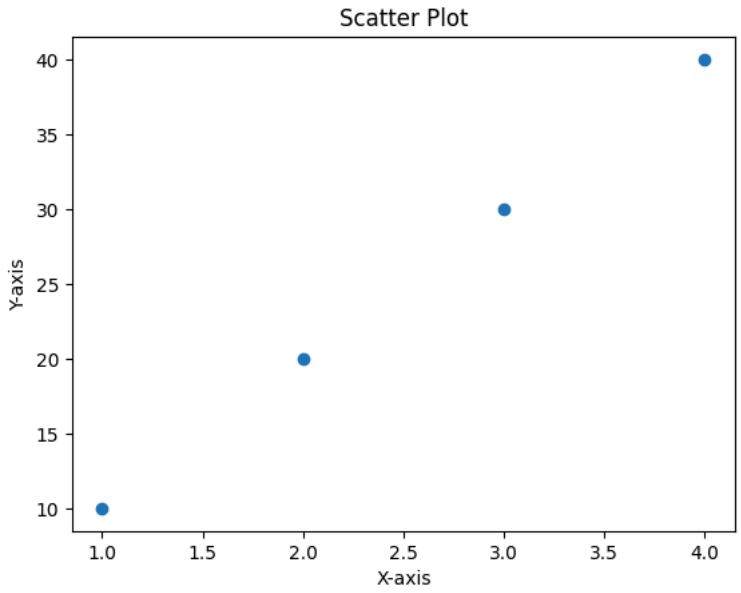
```
#Line Plot
plt.plot(x,y)
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Plot')
plt.show()
```




```
#Line Bar
plt.bar(x,y)
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Plot')
plt.show()
```




```
#Scatter plot
plt.scatter(x,y)
plt.title('Scatter Plot')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.show()
```

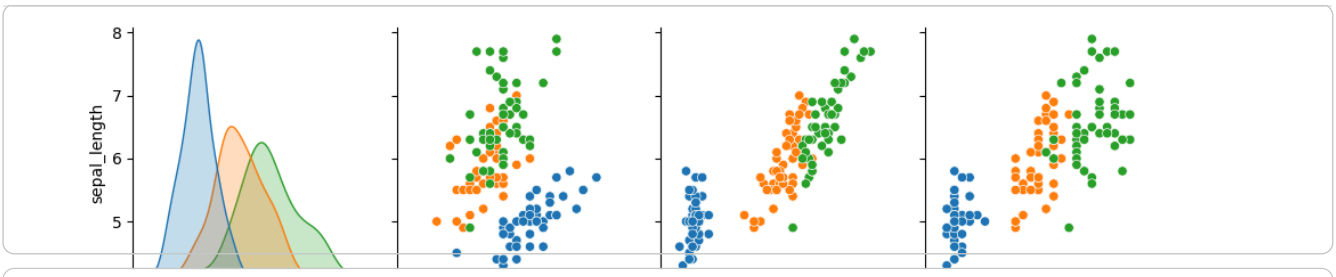


Seaborn

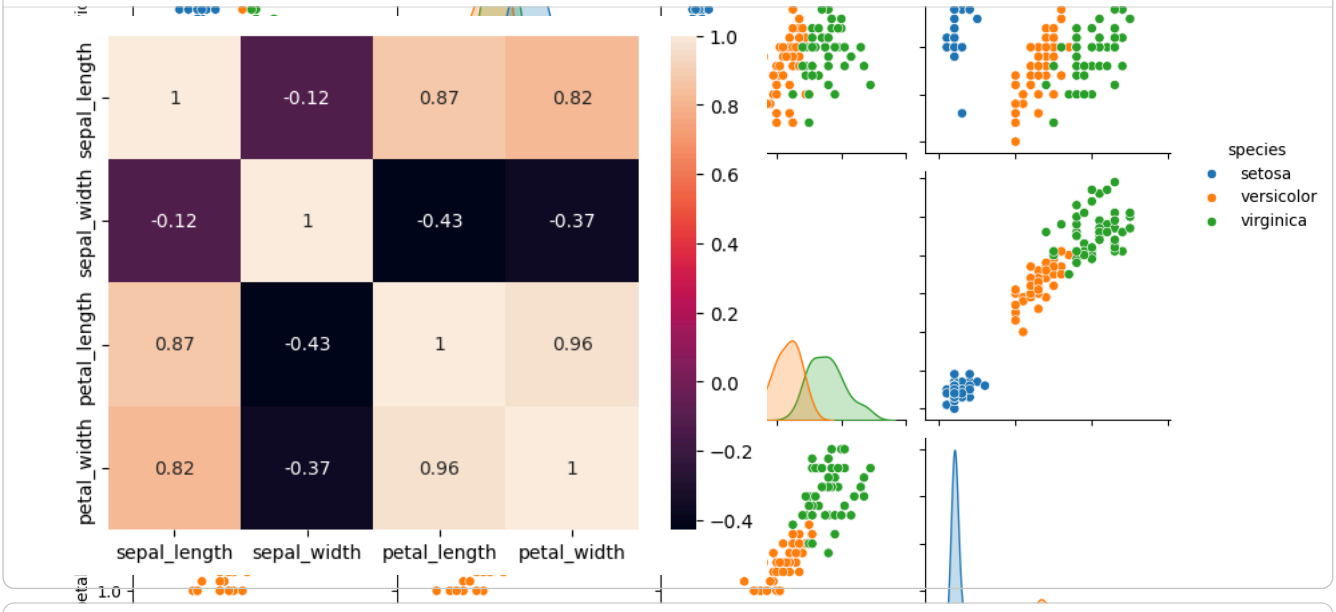
```
import seaborn as sns
import matplotlib.pyplot as plt
```

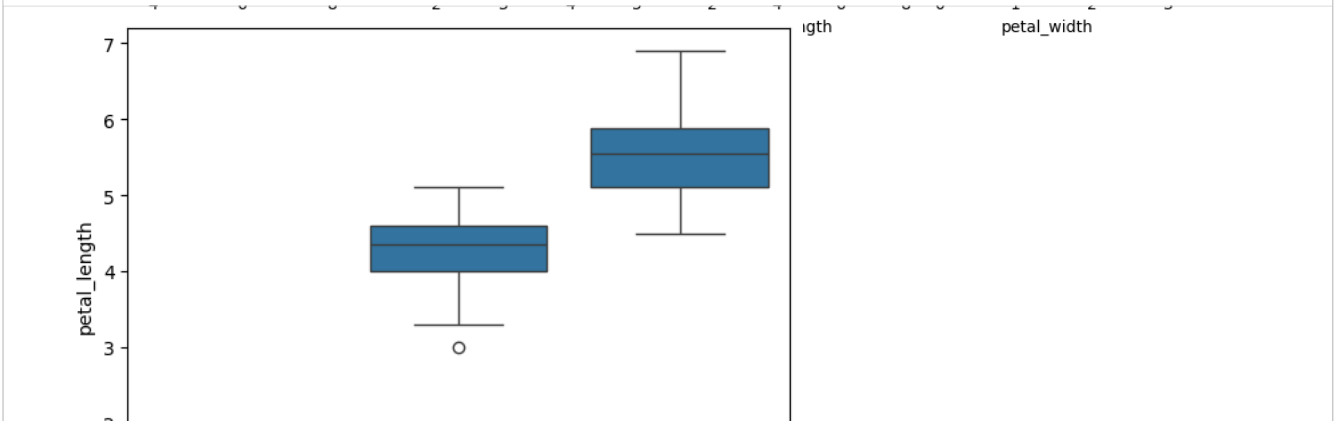
```
#Load sample dataset
data = sns.load_dataset("iris")
```

```
#Pair plot
sns.pairplot(data,hue='species')
plt.show()
```




```
#heatmap
corr = data.corr(numeric_only=True)
sns.heatmap(corr,annot=True)
plt.show()
```




```
#Box plot
sns.boxplot(x='species',y='petal_length',data=data)
plt.show()
```